Article

Framing Effect in the Trolley Problem and Footbridge Dilemma: Number of Saved Lives Matters

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Fei Cao
Department of Psychology, Fourth Military Medical University, Xi'an, China

Jiaxi Zhang
Xi'an Research Institute of High-Technology, Xi'an, China

Lei Song and Shoupeng Wang
Military Economics Academy, Wuhan, China

Danmin Miao
Department of Psychology, Fourth Military Medical University, Xi'an, China

Jiaxi Peng
Military Economics Academy, Wuhan, China; Department of Psychology, Fourth Military Medical University, Xi'an, China

Abstract

The present study investigated the effect of dilemma type, framing, and number of saved lives on moral decision making. A total of 591 undergraduates, with a mean age of 20.56 (SD = 1.37) were randomly assigned to 12 groups on the basis of a grid of two dilemma types (the trolley problem or the footbridge dilemma) by three frames (positive, neutral, or negative frame) by two different numbers of workers (5 or 15 people). The main effects of dilemma type, frame, and number of saved workers were all significant. The interaction of dilemma type and number of saved workers and the interaction of the three independent factors were significant. Results indicated that moral judgment is affected by framing. Specifically, people were more inclined to

Corresponding Authors:
Jiaxi Peng, Military Economics Academy, No. 122 Luojiadun, Wuhan 430035, China.
Email: pengjx880124@hotmail.com

Danmin Miao, Department of Psychology, Fourth Military Medical University, No. 169 Changle West Road, Xi'an 710032, China.
Email: miaodanmin@126.com

utilitarianism in the positive or neutral frame and more inclined to intuitionism in the negative frame. Furthermore, this effect can be moderated by dilemma type and number of saved lives. Implications of our results are discussed.

Keywords
Moral dilemma, framing effect, dilemma types, number of saved lives

Introduction

People encounter various moral issues that involve decision making every day. In recent years, with the increasing emphasis on framing effects in decision making, those involved in moral decision making have become an important topic of investigation (Crockett, Kurth-Nelson, Siegel, Dayan, & Dolan, 2014). The cognitive processing of moral judgments and the factors that influence these judgments have long puzzled philosophers and psychologists (Conway & Gawronski, 2013; Youssef et al., 2012). In early studies of moral decision making, moral judgment was regarded as a pure logical reasoning process (Bargh & Ferguson, 2000; Galotti, 1989). Psychological elements, such as those pertaining to intuition and emotion, were thought to be uninvolved in the process of moral judgment (Bienengräber, 2014). However, this idea has recently been challenged, particularly by cognitive neuroscientists (Crockett et al., 2015). In fact, Moll, Eslinger, and Oliveira-Souza (2001) determined that moral decision making activates brain regions associated with emotion. Several researchers have considered emotion to be a direct factor that actuates people's moral judgments, and that moral judgments originating from instant negative emotions to malicious harm may keep people from immoral acts (Conway & Gawronski, 2013; Phelps, Lempert, & Sokol-Hessner, 2014). Haidt (2001) further proposed that moral judgment is an outcome of rational and irrational processing.

The varying perspectives on moral judgments have coalesced into two schools of thought regarding moral decision making, namely, utilitarianism and intuitionism. Utilitarianism emphasizes the maximum benefits of a behavior without considering other restrictions, as long as the most benefit can be obtained for the most people in the end (Lucas & Livingston, 2014). By contrast, intuitionism emphasizes moral supremacy, which requires people to follow specific rules that constrain certain behaviors (Cubitt, Drouvelis, Gächter, & Kabalin, 2011; Rawls, 2009). These opposing philosophies can predict different outcomes even when individuals are faced with the same social phenomena. However, both schools of thought emphasize the avoidance of harming the innocent. In summary, utilitarianism focuses on benefiting the most people, while intuitionism emphasizes the moral doctrine against violations under any condition.

Moral dilemmas and dual-process theory

Traditionally, theorists have studied the moral decision process using some classic moral dilemmas, such as Haidt's trolley problem, the footbridge dilemma, the prisoner's dilemma, and others (Greene, Nystrom, Engell, Darley, & Cohen, 2004). Haidt describes the following scenario in the trolley problem: Suppose you see a runaway trolley carrier on the tracks. Meanwhile, five workers are repairing the track just in front of it without noticing the danger. The only way to save them is to pull a lever and send the trolley onto another track. However, another worker would be killed by the trolley. Would you pull the lever? Most people would choose to pull the lever and sacrifice one individual to save five others (Haidt, 2001). The circumstance changes in the footbridge dilemma: Suppose you are on a footbridge and see a runaway trolley carrier on the tracks. Meanwhile, five workers are repairing the track just in front of it without noticing the danger. The only way to save them is to push the rather large gentleman beside you onto the tracks and let his body stop the trolley, which would crush him to death. Will you push him or not? In this situation, few people would push the man to save more people (Bazerman & Greene, 2010). The question of whether to sacrifice one person for five will now provoke different moral judgments because of changed circumstances. Utilitarianism states that sacrificing one individual for five is morally acceptable, whereas intuitionism asserts no harm to any innocent individual through fair and foul (Bazerman & Gino, 2012; Strohminger, Lewis, & Meyer, 2011). Evans (2010) proposed a dual-process theory of moral judgment which suggests that people's moral judgment is a concurrent outcome of utilitarianism and intuitionism. Intuitionism would be activated by automatic emotion processing, whereas utilitarianism would be driven by cognitive processing (Evans & Stanovich, 2013; Greene, 2009). Individuals would follow the principle that has a stronger driving force in the competition of these two principles (Moore, Clark, & Kane, 2008). In the trolley problem, pulling the lever that will lead to one worker's death requires a cognitive assessment process, in which people are willing to sacrifice one person to save five people, thus obeying the utilitarianism rule. However, pushing the large gentleman onto the tracks to stop the trolley has a high level of emotional arousal, which adheres to intuitionism (Klein, 2011; Lapsley & Hill, 2008). Dual-process theory states that the strength of utilitarianism could be manipulated by the number of workers to be saved if the level of emotional arousal could be varied using differing scenarios to alter people's moral decisions (Trémolière, De Neys, & Bonnefon, 2012). Thus, this study first hypothesized that the number of workers to be saved could influence individuals' moral decisions. That is, the more people to be saved, the more likely utilitarianism would predominate. Thus, deciding to sacrifice one life for more than five lives would be easier than deciding to sacrifice one life for up to five lives.

Framing effect in decision making

People's decision making and preferences are affected by how information is presented (Peng et al., 2013; Peng, Xiao, Yang, Wu, & Miao, 2014). The framing effect is a phenomenon which occurs when the same problem is presented with different representations of information (or frames). People then make different judgments and decisions according to the information with which the problem is framed (Peng et al., 2014; Zhao, Huang, Li, Zhao, & Peng, 2015). This phenomenon is similar to that which occurs in the "Asian disease problem." In this classic dilemma, the United States' government is preparing to respond to the outbreak of an Asian disease which could result in 600 deaths. Two options are available: Following plan A would save (positive frame) 200 people (400 would die, negative frame), whereas following plan B leads to a one-third possibility of saving (positive frame) all 600 people (nobody dies, negative frame) and a two-third possibility of saving no one (all die, negative frame). Using different frames to describe the same outcomes leads to totally different decisions, in that people tend to be conservative in the positive frame and risky in the negative frame (Tversky & Kahneman, 1981). The framing effect has been considered an important factor in bias in human reasoning (Bruine de Bruin, Parker, & Fischhoff, 2007). Studies have shown that the framing effect is a stable and robust phenomena in economics, medicine, and management decision making (Levin, Gaeth, Schreiber, & Lauriola, 2002; Peng, Jiang, Miao, Li, & Xiao, 2013).

However, research on the framing effect has focused mainly on risk decisions. Few studies have investigated how information descriptions can affect moral decision making. For example, will the descriptions affect people's moral judgment if the "to be saved" (positive) frame and the "to be killed" (negative) frame are used to describe the dilemma, which are similar to "pulling the lever, one worker will be killed; otherwise, five workers will be killed on the main tracks" or "pulling the lever, five workers will be saved on the main tracks; otherwise, one worker will be saved," respectively? Petrinovich and O'Neill (1996) conducted a preliminary exploration into this question and determined that the negative frame had a significant effect on moral decision making, whereas both the positive and neutral frames exerted far less influence on simplex situations. Broeders, Van Den Bos, Müller, and Ham (2011) determined that the number of participants who expressed "to be saved" and decided to push the large man onto the tracks was significantly greater than those who expressed "to be saved" in the footbridge dilemma. However, this finding showed no significant differences when the same frames were couched in the trolley problem: more people chose to pull the lever to save more people. The authors emphasized that framing effects on final decisions existed even when participants did not realize the descriptions had been changed in the experiment (Broeders et al., 2011). However, few studies have pondered the framing effect in moral decision making. First, the explanations of the framing effect given by Broeders' studies are unclear and did not provide convincing evidence. Further, existing research on moral judgment mostly focuses on Western participants. The external validity of the framing effect in moral decision making in Eastern cultures needs to be explored, as evidence suggests that the two cultures differ in the cognitive assessment processes of decision making (Hamamura, 2012). Thus, this study tests in an Eastern culture, the hypothesis that different framing descriptions will bring about disparate moral judgments, and indicate a framing effect in moral decision making.

Method

Participants

Participants were 591 undergraduates (294 men, 297 women, $M_{age} = 20.56$ years, age range: 17–22 years) who majored in English, physics, and history, at three universities in China. Of the 591 inventories distributed, 570 were valid and 21 were blank, which constituted a response rate of 96.44%.

Materials

The research material was developed from Haidt's (2001) trolley problem and footbridge dilemma. Some adjustments were made to the original dilemma in order to manipulate the frames and the number of workers to be saved. Each participant had to decide whether he or she would take the relevant actions or not, and indicate this response with a rating on a 6-point Likert scale.

The dilemma is posed to the participants as follows: Suppose you see a running tram on the main track. On the same track ahead, 5/15 workers are repairing the track and unable to move as the trolley heads straight for them. The only thing you can do is to pull a lever to switch to a different track, on which there is only one person. Five (15) workers will be saved if you pull the lever. One worker on another track will be saved if you do not pull the lever (positive frame). You must rate the strength of your decision on a scale from 1 (definitely do not pull the lever) to 6 (definitely pull the lever). As shown in Table 1, the situation was described in 12 different conditions, that is, 2 dilemma types (the trolley problem or the footbridge dilemma) $\times$ 3 frames (positive, neutral, or negative frame) $\times$ 2 numbers of workers (5 or 15 people).

Procedure

Participants were randomly assigned to one of the 12 on the basis of a grid of 2 dilemma types (the trolley problem or the footbridge dilemma) $\times$ 3 frames (positive, neutral, or negative frame) $\times$ 2 numbers of workers (5 or 15 people)

Table 1. Research materials.

	Dilemma type	Positive frame	Neutral frame	Negative frame
Trolley problem	Suppose you see a running tram on the main road, and 5 (15) workers are repairing the track in front of you. If no actions are taken, then the 5 workers will die. You can only to save the 5 workers if you pull a lever to switch tracks, which will kill one worker on another track.	5 (15) workers will be saved if you pull the lever; one worker on another track will be saved if you do not pull the lever. All your actions are legal and understandable. Will you pull the lever?	5 (15) workers will be saved if you pull the lever, but another worker will die. One worker will be saved if you do not pull the lever, but 5 workers will die. All your actions are legal and understandable. Will you pull the lever?	One worker will die if you pull the lever. 5 (15) workers will die if you do not pull the lever. All your actions are legal and understandable. Will you pull the lever?
Footbridge dilemma	Suppose you are standing on a bridge and see a runaway tram with no fork on track. 5 (15) workers will be killed on the front rail if no actions are taken. You are standing next to a large man. The only way to save the 5 workers is if you push the large man under the bridge. The man will be killed, but his body will stop the trolley and save the workers.	5 (15) workers will be saved if you push the large man; the large man will be saved if you do not push him. All your actions are legal and understandable. Will you push the large man?	5 (15) workers will be saved if you push the large man and let him die. If you do not push the large man, then the 5 (15) workers will die. All your actions are legal and understandable. Will you push the large man?	The large man will die if you push him. 5 (15) workers will be killed if you do not push the large man. All your actions are legal and understandable. Will you push the large man?

Table 2. Descriptive statistics in all groups (mean ± SD).

Dilemma type	Saved lives	Positive frame	Neutral frame	Negative frame
Trolley problem	5	5.04 ± 1.11	4.53 ± 1.70	4.19 ± 1.70
	15	4.91 ± 1.46	5.06 ± 0.99	5.00 ± 1.34
Footbridge problem	5	3.17 ± 1.72	2.85 ± 1.91	3.09 ± 1.75
	15	4.31 ± 1.50	4.30 ± 1.33	3.37 ± 1.53

Table 3. The main and interaction effects of dilemma type, frames, and saved lives on moral decision making.

	F	*df*	p	Effect size (Partial η^2)
Dilemma type	99.786**	1	<0.01	0.152
Saved lives	28.520**	1	<0.01	0.049
Frame	4.176*	2	0.02	0.015
Dilemma type × saved lives	4.705*	1	0.03	0.008
Dilemma type × frame	0.127	2	0.88	<0.001
Saved lives × frame	1.444	2	0.24	0.005
Dilemma type × frame × saved lives	4.639*	2	0.01	0.016

Note: *$p < 0.05$; **$p < 0.01$.

adopting a between-subject design. Participants responded to the decision making problem were ranked by adopting a 6-point Likert scale. A 6-point scale was used, instead of the more common 7-point scale, in order to require participants to express a preference for one action over the other by denying them a mid-point choice (Xiao et al., 2015). Moreover, the 6-point scale helped determine the strength of the choice made by the participants (Peng, He, et al., 2013). A large value indicated that the participants were prone to utilitarianism. All participants were informed that they would take part in a psychological investigation, their responses were anonymous, and that each question had no right or wrong answer. The 12 groups of participants were not allowed to communicate until they completed the questionnaires to avoid errors caused by comparisons among groups.

Results

The main effects of dilemma type, frame, and numbers of saved workers (5 or 15) were assessed using a 2 × 3 × 2 ANOVA (means and SD in each cell are shown in Table 2). The results presented in Table 3 indicate that the main effect

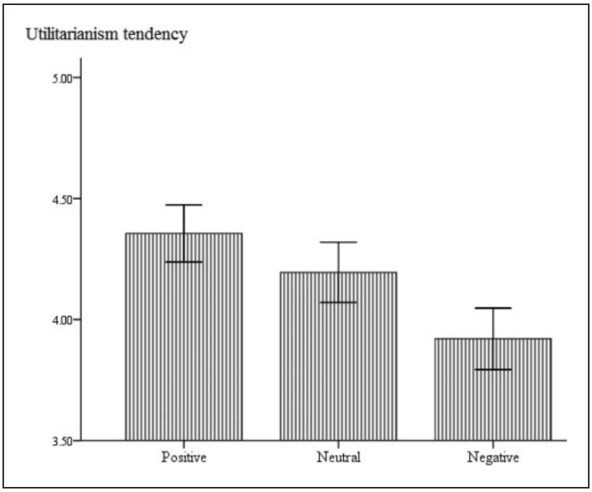

Figure 1. Framing effect in moral decision making.

of dilemma type ($F_{1,570} = 99.786$, $p < 0.01$), frame ($F_{2,570} = 4.176$, $p = 0.016$), and numbers of saved workers ($F_{1,570} = 28.52$, $p < 0.01$) were significant. Moreover, the interaction between dilemma type and saved workers ($F_{1,570} = 4.705$, $p = 0.03$) and the interaction of the three independent variables ($F_{2,570} = 4.639$, $p = 0.01$) were significant.

Figure 1 shows a bar graph depicting the results of a simple effects analysis of the number of saved workers in each frame, in both the trolley problem and the footbridge dilemma. The results indicate that people are more willing to take action to save more workers in the positive or neutral frame than in the negative frame ($F_{2,570} = 3.189$, $p = 0.042$). In other words, people are more inclined to utilitarianism in the positive or neutral frame and more inclined to intuitionism in the negative frame. Thus, a significant framing effect in moral decision making was detected here.

This study separately tested the framing effect with dilemma type and number of saved lives. Figure 2 shows these results. In the trolley problem, the framing effect was significant ($F = 3.80$, $p = 0.03$) when the number of saved lives was 5 but was not significant ($F = 0.39$, $p = 0.68$) when the number was 15. In the footbridge dilemma, the framing effect was not significant ($F = 0.16$, $p = 0.85$) when the number of workers to be saved was 5 but was significant ($F = 6.39$, $p < 0.01$) when the number of saved lives were 15.

Discussion

This study investigated the influence of different framing descriptions and the number of workers to be saved, on moral decision making in different moral

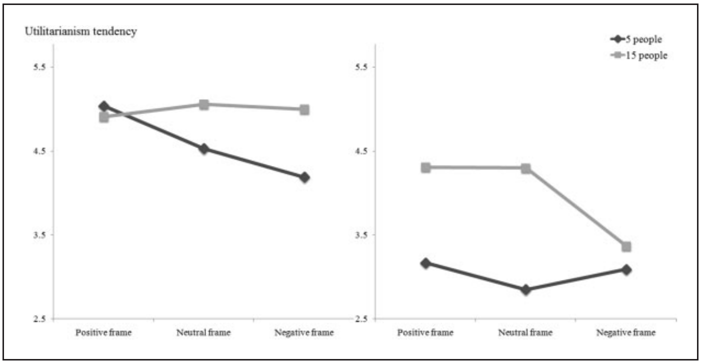

Figure 2. Framing effect within dilemma types and saved lives.

dilemmas. The main effect of number of saved workers was significant. Specifically, people intend to take action more when the number of workers to be saved increases. That is, more people would use the utilitarianism principle in their moral decision making as the benefit increases. This finding confirms previous discoveries. Based on the dual-process model of moral judgment, utilitarian behaviors driven by cognitive processing would increase if the number of workers to be saved increases, thus dominating in the competition between it and intuitionist judgments driven by emotional processing (Greene, 2007, 2009). The significance of the current study is that it provides for the empirical validity of previous studies and shows that utilitarian reasoning plays a role in moral decision making of members of more collectivist Eastern cultures.

Another finding of the current study was that a significant framing effect is observed in moral decisions. This was illustrated by the significant main effect of frame and interaction between dilemma type and number of saved workers. The simple effects analysis further showed that people are likely to take actions and choose the alternative that maximizes the benefits for the most people under the positive and neutral frames. In other words, people lean toward utilitarianism when the decision making dilemma is described in nonnegative terms. However, people are likely to do nothing and lean toward intuitionism when the dilemmas are described using a negative frame. Hume (1740) believed that individuals' moral judgments originate from instant negative emotions to malicious harms, and people will not go against the intuitionism principle to avoid negative emotions. People have less negative emotions when moral dilemmas are described in terms of lives "saved" versus lives "killed." Therefore, the intuitionism principle driven by emotion preponderates in judgments and makes decision makers elude or choose to do nothing. Descriptions using the positive frame emphasize how many people can be saved owing to decision makers' behaviors. This leads the decision makers to readily compare the outcomes as to the amount and lean toward choosing to save more people (Deppe et al., 2005). However, using the negative frame, people realize that "one innocent worker/person will die because of my behavior." They attribute the loss to their acts, which causes them severe mental pain, and thus, decide to do nothing (Nabi, 2003).

Moreover, the current study also determined that whether the framing effect was significant or not is determined by dilemma type and number of saved workers. In other words, the framing effect in moral decision making was influenced by the moral situation and number of saved lives. It can be said that in the trolley problem, the framing effect was significant only when the number of saved workers was relatively small. However, in the footbridge dilemma, the framing effect was significant only when the number of saved workers was large. This phenomenon can be explained by the comparatively small effect size of framing descriptions (Evans & Curtis-Holmes, 2005). According to the dual-process model of moral decision making, in the trolley problem, utilitarian judgment has a significant advantage over intuitionism. Therefore, people are prone to obey the utilitarian principle regardless of how the problem is described. However, the process of utilitarianism does not have an advantage over intuitionism when the number of workers to be saved is five. Thus, the framing effect is significant. In the footbridge dilemma, pushing the large gentleman onto the tracks when the number of workers to be saved is five would cause severe negative emotions, encouraging people's intuitionist behavior. Therefore, people are prone to obey the intuitionism principle regardless of how the problem is described (Wixted, 2007). Moral judgments are readily altered by framing descriptions when the number of workers increases to 15. This change leads to a significant framing effect in the condition.

The implications of this research are many. For instance, the findings may inform planning and decision-making pertaining to a community's or nation's civil and legal matters. In this regard, our research makes a valuable contribution to understanding moral problem solving from an Eastern cultural perspective. More research differentiating cultural variance in responding will further elucidate relevant differences between nations and perhaps the rationale buttressing their policies.

There are a few limitations of the current study which should be pointed out. First, moral judgment is often tested in the form of moral dilemmas. We utilized this methodology in the present study instead of using empirical studies, due to ethical restrictions and the difficulty of creating real experimental conditions. However, such a methodology has unavoidable limitations. For instance, decisions made in reality may be not consistent with those that could be triggered by hypothetical dilemmas. Second, the results of the present study rely on previous research that can help inform the interpretation of the outcomes. Conducting direct research on moral decisions based on neurophysiology remains a formidable challenge for practical as well as ethical reasons. However, addressing these limitations to the extent possible and clarifying the relationships among moral dilemma types, frames, benefit level (number of saved lives), and moral decision making are still important endeavors. Third, the current study adopts the method of paper-pencil testing to measure the participants' moral decision making. Other indicators, such as reaction time, which helps to distinguish whether participants will utilize utilitarian or intuitionist principles in moral decision making, are very difficult to collect. Thus, future studies will benefit by adopting computer-based testing to get more objective indicators of people's moral decision making. One last limitation of the current study is that we necessarily included a sentence in the research materials telling participants that ''all your actions are legal and understandable.'' Given Chinese law, this statement was absolutely necessary to ensure that the moral dilemma was candidly answered. However, this inclusion may also have lead the participants somewhat.

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References

Bargh, J. A., & Ferguson, M. J. (2000). Beyond behaviorism: On the automaticity of higher mental processes. *Psychological Bulletin*, *126*(6), 925.

Bazerman, M. H., & Gino, F. (2012). Behavioral ethics: Toward a deeper understanding of moral judgment and dishonesty. *Annual Review of Law and Social Science*, *8*, 85–104.

Bazerman, M. H., & Greene, J. D. (2010). In favor of clear thinking: Incorporating moral rules into a wise cost-benefit analysis—Commentary on Bennis, Medin, & Bartels (2010). *Perspectives on Psychological Science*, *5*(2), 209–212.

Bienengräber, T. (2014). Moral development in business education–social conditions influencing moral judgement competence. *Journal of Vocational Education & Training*, *66*(3), 406–427.

Broeders, R., Van Den Bos, K., Müller, P. A., & Ham, J. (2011). Should I save or should I not kill? How people solve moral dilemmas depends on which rule is most accessible. *Journal of Experimental Social Psychology*, *47*(5), 923–934.

Bruine de Bruin, W., Parker, A. M., & Fischhoff, B. (2007). Individual differences in adult decision-making competence. *Journal of Personality and Social Psychology*, *92*(5), 938.

Conway, P., & Gawronski, B. (2013). Deontological and utilitarian inclinations in moral decision making: A process dissociation approach. *Journal of Personality and Social Psychology*, *104*(2), 216.

Crockett, M. J., Kurth-Nelson, Z., Siegel, J. Z., Dayan, P., & Dolan, R. J. (2014). Harm to others outweighs harm to self in moral decision making. *Proceedings of the National Academy of Sciences*, *111*(48), 17320–17325.

Crockett, M. J., Siegel, J. Z., Kurth-Nelson, Z., Ousdal, O. T., Story, G., Frieband, C., & Dolan, R. J. (2015). Dissociable effects of serotonin and dopamine on the valuation of harm in moral decision making. *Current Biology*, *25*(14), 1852–1859.

Cubitt, R. P., Drouvelis, M., Gächter, S., & Kabalin, R. (2011). Moral judgments in social dilemmas: How bad is free riding? *Journal of Public Economics*, *95*(3), 253–264.

Deppe, M., Schwindt, W., Kraemer, J., Kugel, H., Plassmann, H., Kenning, P., & Ringelstein, E. B. (2005). Evidence for a neural correlate of a framing effect: Bias-specific activity in the ventromedial prefrontal cortex during credibility judgments. *Brain Research Bulletin*, *67*(5), 413–421.

Evans, J. S. B., & Curtis-Holmes, J. (2005). Rapid responding increases belief bias: Evidence for the dual-process theory of reasoning. *Thinking & Reasoning*, *11*(4), 382–389.

Evans, J. S. B. (2010). Intuition and reasoning: A dual-process perspective. *Psychological Inquiry*, *21*(4), 313–326.

Evans, J. S. B., & Stanovich, K. E. (2013). Dual-process theories of higher cognition advancing the debate. *Perspectives on Psychological Science*, *8*(3), 223–241.

Galotti, K. M. (1989). Approaches to studying formal and everyday reasoning. *Psychological Bulletin*, *105*(3), 331.

Greene, J. D. (2007). Why are VMPFC patients more utilitarian? A dual-process theory of moral judgment explains. *Update*, *11*(8), 322–323.

Greene, J. D. (2009). Dual-process morality and the personal/impersonal distinction: A reply to McGuire, Langdon, Coltheart, and Mackenzie. *Journal of Experimental Social Psychology*, *45*(3), 581–584.

Greene, J. D., Nystrom, L. E., Engell, A. D., Darley, J. M., & Cohen, J. D. (2004). The neural bases of cognitive conflict and control in moral judgment. *Neuron*, *44*(2), 389–400.

Haidt, J. (2001). The emotional dog and its rational tail: A social intuitionist approach to moral judgment. *Psychological Review*, *108*(4), 814–834. doi: 10.1037/0033-295x.108.4.814

Hamamura, T. (2012). Are cultures becoming individualistic? A cross-temporal comparison of individualism–collectivism in the United States and Japan. *Personality and Social Psychology Review*, *16*(1), 3–24.

Hume, D. (1740). *A Treatise Concerning Human Understanding*. London: Clarendon.

Klein, C. (2011). The dual track theory of moral decision-making: a critique of the neuroimaging evidence. *Neuroethics*, *4*(2), 143–162.

Lapsley, D. K., & Hill, P. L. (2008). On dual processing and heuristic approaches to moral cognition. *Journal of Moral Education*, *37*(3), 313–332.

Levin, I. P., Gaeth, G. J., Schreiber, J., & Lauriola, M. (2002). A new look at framing effects: Distribution of effect sizes, individual differences, and independence of types of effects. *Organizational Behavior and Human Decision Processes*, *88*(1), 411–429.

Lucas, B. J., & Livingston, R. W. (2014). Feeling socially connected increases utilitarian choices in moral dilemmas. *Journal of Experimental Social Psychology*, *53*, 1–4.

Moll, J., Eslinger, P. J., & Oliveira-Souza, R. D. (2001). Frontopolar and anterior temporal cortex activation in a moral judgment task: Preliminary functional MRI results in normal subjects. *Arquivos de Neuro-Psiquiatria*, *59*(3B), 657–664.

Moore, A. B., Clark, B. A., & Kane, M. J. (2008). Who shalt not kill? Individual differences in working memory capacity, executive control, and moral judgment. *Psychological Science*, *19*(6), 549–557.

Nabi, R. L. (2003). Exploring the framing effects of emotion do discrete emotions differentially influence information accessibility, information seeking, and policy preference? *Communication Research*, *30*(2), 224–247.
Peng, J., He, F., Zhang, Y., Liu, Q., Miao, D., & Xiao, W. (2013). Differences in simulated doctor and patient medical decision making: A construal level perspective. *PLoS ONE*, *8*(11), e79181.
Peng, J., Jiang, Y., Miao, D., Li, R., & Xiao, W. (2013). Framing effects in medical situations: Distinctions of attribute, goal and risky choice frames. *Journal of International Medical Research*, *41*(3), 771–776.
Peng, J., Xiao, W., Yang, Y., Wu, S., & Miao, D. (2014). The impact of trait anxiety on self-frame and decision making. *Journal of Behavioral Decision Making*, *27*(1), 11–19.
Petrinovich, L., & O'Neill, P. (1996). Influence of wording and framing effects on moral intuitions. *Ethology and Sociobiology*, *17*(3), 145–171.
Phelps, E. A., Lempert, K. M., & Sokol-Hessner, P. (2014). Emotion and decision making: multiple modulatory neural circuits. *Annual Review of Neuroscience*, *37*, 263–287.
Rawls, J. (2009). *A theory of justice*. Boston: Harvard University Press.
Strohminger, N., Lewis, R. L., & Meyer, D. E. (2011). Divergent effects of different positive emotions on moral judgment. *Cognition*, *119*(2), 295–300.
Trémolière, B., De Neys, W., & Bonnefon, J.-F. (2012). Mortality salience and morality: Thinking about death makes people less utilitarian. *Cognition*, *124*(3), 379–384.
Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, *211*(4481), 453–458.
Wixted, J. T. (2007). Dual-process theory and signal-detection theory of recognition memory. *Psychological Review*, *114*(1), 152.
Xiao, W., Wu, Q., Yang, Q., Zhou, L., Jiang, Y., Zhang, J., & Peng, J. (2015). Moral Hypocrisy on the basis of construal level: To be a utilitarian personal decision maker or to be a moral advisor? *PLoS ONE*, *10*(2), e0117540. doi: 10.1371/journal.pone.0117540
Youssef, F. F., Dookeeram, K., Basdeo, V., Francis, E., Doman, M., Mamed, D., & Ditshotlo, P. (2012). Stress alters personal moral decision making. *Psychoneuroendocrinology*, *37*(4), 491–498.
Zhao, X., Huang, C., Li, X., Zhao, X., & Peng, J. (2015). Dispositional optimism, self-framing and medical decision-making. *International Journal of Psychology*, *50*(2), 121–127.

Author Biographies

Fei Cao is a PhD candidate at the department of psychology, Fourth Military Medical University. Her research focuses on attitude change and persuasion.

Jiaxi Zhang is an assistant professor of psychology at Xi'an Research Institute of High-Technology. His research focuses on behavioral decision making and positive psychology.

Lei Song is a professor of psychology at Military Economics Academy. Her research focuses on mental health and youth psychology.

Shoupeng Wang is an assistant professor at Military Economics Academy. His research focuses on sports and health.

Danmin Miao is a professor of psychology at Fourth Military Medical University. His research focuses on personnel selection and decision making capacity measurement.

Jiaxi Peng is an assistant professor of psychology at Military Economics Academy. His research focuses on behavioral decision making, especially on framing effect and reference point adoption.